F18CF - Assignment 1

Instructions:

- This assignment is due at 23:59 (Edinburgh Time) on the date specified above
- Your solutions can be written or typeset. No paper submission is necessary
- *Gradescope* submission link available on assignment page in Canvas
- The assessment criteria are listed on the assignment page
- This assignment only requires results in Chapters 1 and 2 from the notes
- Start each question on a new page (Parts of the same question can be on the same page).
- This assignment contains 4 questions worth a total of 40 marks. Marks are listed with each question. An additional 3 marks are awarded for meeting the submission standards.

Questions:

9 marks 1. (a) Express $\begin{pmatrix}-1\\10\\-9\end{pmatrix}$ as a linear combination of $\begin{pmatrix}1\\4\\-1\end{pmatrix}$ and $\begin{pmatrix}3\\5\\2\end{pmatrix}$.

Solution: Construct a linear combination equation.

$$\begin{pmatrix}-1\\10\\-9\end{pmatrix} = \alpha\begin{pmatrix}1\\4\\-1\end{pmatrix} + \beta\begin{pmatrix}3\\5\\2\end{pmatrix}$$

Writing in matrix forms gives

$$\begin{pmatrix}1 & 3\\4 & 5\\-1 & 2\end{pmatrix}\begin{pmatrix}\alpha\\\beta\end{pmatrix} = \begin{pmatrix}-1\\10\\-9\end{pmatrix}$$

and augmented form (with row reductions) give:

$$\left(\begin{array}{cc|c}1 & 3 & -1\\4 & 5 & 10\\-1 & 2 & -9\end{array}\right) \sim \left(\begin{array}{cc|c}1 & 3 & -1\\0 & -7 & 14\\0 & 5 & -10\end{array}\right) \begin{array}{l}R_2 \to R_2 - 4R_1\\R_3 \to R_3 + R_1\end{array}$$

$$\sim \left(\begin{array}{cc|c}1 & 3 & -1\\0 & -7 & 14\\0 & 0 & 0\end{array}\right) \quad R_3 \to R_3 + \tfrac{5}{7}R_2$$

Back substitutions give $\beta = -2$ and hence $\alpha = 5$.

(b) Determine if the following set of vectors form a linearly independent set in $\mathbb{R}^3$

$$\left\{\begin{pmatrix}2\\-2\\3\end{pmatrix}, \begin{pmatrix}3\\0\\3\end{pmatrix}, \begin{pmatrix}0\\5\\1\end{pmatrix}\right\}$$

Solution: Construct a linear combination equation.

$$\begin{pmatrix}0\\0\\0\end{pmatrix} = \alpha\begin{pmatrix}2\\-2\\3\end{pmatrix} + \beta\begin{pmatrix}3\\0\\3\end{pmatrix} + \gamma\begin{pmatrix}0\\5\\1\end{pmatrix}$$

Writing in matrix forms gives

$$\begin{pmatrix} 2 & 3 & 0 \\ -2 & 0 & 5 \\ 3 & 3 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

and augmented form (with row reductions) give:

$$\left(\begin{array}{ccc|c} 2 & 3 & 0 & 0 \\ -2 & 0 & 5 & 0 \\ 3 & 3 & 1 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 2 & 3 & 0 & 0 \\ 0 & 3 & 5 & 0 \\ 0 & -\frac{3}{2} & 1 & 0 \end{array}\right) \begin{array}{l} R_2 \to R_2 + R_1 \\ R_3 \to R_3 - \frac{3}{2}R_1 \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 2 & 3 & 0 & 0 \\ 0 & 3 & 5 & 0 \\ 0 & 0 & \frac{7}{2} & 0 \end{array}\right) \quad R_3 \to R_3 + \tfrac{1}{2}R_2$$

With a full set of pivots, There is a unique solution

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

and we can conclude that $\alpha = \beta = \gamma = 0$. Hence the set of vectors are linearly independent.

(c) Let $x, y, z \in \mathbb{R}$ and define

$$A = \begin{pmatrix} x & 0 & y \\ 0 & x-2 & 0 \\ x & z+1 & x \end{pmatrix}$$

Compute $\det(A)$ and determine the conditions on x, y, z such that $\det(A) > 0$.

Solution: We want to expand along the row/columns with most zeros to simplify our work. In this case, we pick the row 2.

$$\begin{aligned} \det(A) &= \det \begin{pmatrix} x & 0 & y \\ 0 & x-2 & 0 \\ x & z+1 & x \end{pmatrix} \\ &= (x-2) \cdot \det \begin{pmatrix} x & y \\ x & x \end{pmatrix} \\ &= (x-2)(x^2 - xy) = x(x-2)(x-y) \end{aligned}$$

Since the determinant is independent of z, we have $z \in \mathbb{R}$. We determine the side of the determinant by looking at the sign of $x(x-2)$ and then picking y suitable. This gives two conditions:

- $z \in \mathbb{R}$, $x \in (0, 2)$, and $x < y$. Since $x(x-2) < 0$, we also need $x - y < 0$. Or,
- $z \in \mathbb{R}$, $x \in \mathbb{R} \backslash [0, 2]$, and $x > y$. Since $x(x-2) > 0$, we also need $x - y > 0$

8 marks 2. Consider the matrix defined by

$$M = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ -3 & 0 & 1 \end{pmatrix}$$

(a) Define the vector $\vec{v} = (1, 1, 0)^T$ and compute the vectors $M\vec{v}$ and $M^2\vec{v}$

Solution: Multiplying the matrix by the vector gives:

$$\vec{v} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$M\vec{v} = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ -3 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -3 \end{pmatrix}$$

$$M^2\vec{v} = M(M\vec{v}) = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ -3 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ -9 \end{pmatrix}$$

(b) Show that the set of vectors $\left\{\vec{v}, M\vec{v}, M^2\vec{v}\right\}$ form a basis of $\mathbb{R}^3$

Solution: Using Theorem 2.24, we only have to show that the set of vectors are linear independent. Starting with the linear independent equation:

$$\alpha \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 2 \\ -3 \end{pmatrix} + \gamma \begin{pmatrix} 1 \\ 4 \\ -9 \end{pmatrix} = \vec{0}$$

We can convert into an augmented matrix and apply row reduction

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & 2 & 4 & 0 \\ 0 & -3 & -9 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & -3 & -9 & 0 \end{array}\right) \qquad R_2 \to R_2 - R_1$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & -9 & 0 \\ 0 & 0 & 3 & 0 \end{array}\right) \qquad R_2 \leftrightarrow R_3$$

Since we have the full set of pivots (or back-substituting), we have a unique solution $\alpha = \beta = \gamma = 0$ and hence the vectors are linearly independent.

Solution: (Alternate) Using Theorem 2.26, we can also show the vectors are linearly independent using the determinant.

$$\det \begin{pmatrix} 1 & 2 & 1 \\ 1 & 2 & 4 \\ 0 & -3 & -9 \end{pmatrix}$$

Expand along column 1

$$\det\begin{pmatrix}1 & 2 & 1\\ 1 & 2 & 4\\ 0 & -3 & -9\end{pmatrix} = 1\cdot\det\begin{pmatrix}2 & 4\\ -3 & -9\end{pmatrix} - 1\cdot\det\begin{pmatrix}2 & 1\\ -3 & -9\end{pmatrix}$$
$$= 1\cdot(2\cdot(-9) - 4\cdot(-3)) - 1\cdot(2\cdot(-9) - 1\cdot(-3)) = 9$$

Since the determinant is non-zero, we know that the vectors are linearly independent and hence form a basis by Theorem 2.24.

(c) Let $\vec{u} \in \mathbb{R}^3$ be a non-zero vector. Prove the following statement or construct a counter-example to disprove the statement:

The set $\{\vec{u}, M\vec{u}, M^2\vec{u}\}$ always forms a basis of $\mathbb{R}^3$.

Solution: We can try to create linearly dependent system as a counter-example. In particular, if we have $\vec{u} = M\vec{u}$, then $\vec{u} = M\vec{u} = M^2\vec{u}$ and the set cannot form a basis. We can explicitly construct an example.

$$\begin{pmatrix}0 & 2 & 1\\ 1 & 1 & 0\\ -3 & 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix} = \begin{pmatrix}x\\ y\\ z\end{pmatrix}$$
$$\begin{pmatrix}0-1 & 2 & 1\\ 1 & 1-1 & 0\\ -3 & 0 & 1-1\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix} = \begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

which we can create an augmented matrix and apply row reduction:

$$\left(\begin{array}{ccc|c}-1 & 2 & 1 & 0\\ 1 & 0 & 0 & 0\\ -3 & 0 & 0 & 0\end{array}\right) \sim \left(\begin{array}{ccc|c}1 & 0 & 0 & 0\\ -1 & 2 & 1 & 0\\ -3 & 0 & 0 & 0\end{array}\right) \qquad R_1 \leftrightarrow R_2$$
$$\sim \left(\begin{array}{ccc|c}1 & 0 & 0 & 0\\ 0 & 2 & 1 & 0\\ 0 & 0 & 0 & 0\end{array}\right) \qquad \begin{array}{l}R_2 \to R_2 + R_1\\ R_3 \to R_3 + 3R_1\end{array}$$

This gives $x = 0$ and $z = -2y$. So if $\vec{u}$ is of the form $\vec{u} = \alpha(0, -1, 2)^T$ for any $\alpha \in \mathbb{R}$, we have $\vec{u} = M\vec{u} = M^2\vec{u}$ meaning that the set cannot form a basis.

For completeness: More interesting examples exist where $\vec{u} \neq M\vec{u} \neq M^2\vec{u}$ but the span is not $\mathbb{R}^3$. For example: $v = (1, 0, 0)^T$ gives

$$\vec{u} = \begin{pmatrix}1\\ 0\\ 0\end{pmatrix} \qquad M\vec{u} = \begin{pmatrix}0\\ 1\\ -3\end{pmatrix} \qquad M^2\vec{u} = \begin{pmatrix}-1\\ 1\\ -3\end{pmatrix}$$

which is linearly dependent since $M^2\vec{u} = M\vec{u} - \vec{u}$ and hence cannot form a basis.

11 marks

3. A subset S of $\mathbb{R}^n$ is said to be closed under an operation $*$ on $\mathbb{R}^n$ if for every $s_1, s_2 \in S$, $s_1 * s_2 \in S$.

Let $a \in \mathbb{R}$. In this question, we define a subset U_a of $\mathbb{R}^3$ as

$$U_a = \left\{(x_1, x_2, x_3)^T \in \mathbb{R}^3 \middle| x_1 + 2x_2 + 3ax_3 = a\right\}$$

(a) Determine the possible value(s) of a given that $(2, -3, 3)^T \in U_a$.

Solution: Note that $(2, -3, 3)^T \in U_a$ implies that $2 + 2(-3) + 3a(3) = a$, which gives $a = \frac{1}{2}$.

(b) In the case when $a = 0$, show that U_0 is closed under vector addition.

Solution: If $a = 0$, we have $U_0 = \{(x_1, x_2, x_3)^T \in \mathbb{R}^3 | x_1 + 2x_2 = 0\}$.
Let $\vec{u} = (u_1, u_2, u_3)^T, \vec{v} = (v_1, v_2, v_3)^T \in U_0$. This implies $u_1 + 2u_2 = 0$ and $v_1 + 2v_2 = 0$.
Vector addition gives $\vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3)^T$ which satisfies

$$(u_1 + v_1) + 2(u_2 + v_2) = (u_1 + 2u_2) + (v_1 + 2v_2) = 0 + 0 = 0$$

Hence, $\vec{u} + \vec{v} \in U_0$, that is U_0 is closed under vector addition.

(c) The converse of (b) is the following statement:

If $U_a = \left\{(x_1, x_2, x_3)^T \in \mathbb{R}^3 \middle| x_1 + 2x_2 + 3ax_3 = a\right\}$ is closed under addition, then $a = 0$.

Prove that the above statement is true, or construct a counter-example to disprove it.

Solution: Let $\vec{u} = (u_1, u_2, u_3)^T, \vec{v} = (v_1, v_2, v_3)^T \in U_a$, this implies $u_1 + 2u_2 + 3au_3 = a$ and $v_1 + 2v_2 + 3av_3 = a$. Summing up both equations give:

$$(u_1 + v_1) + 2(u_2 + v_2) + 3a(u_3 + v_3) = 2a$$

From the assumption $\vec{u} + \vec{v} \in U$, gives

$$(u_1 + v_1) + 2(u_2 + v_2) + 3a(u_3 + v_3) = a$$

Equating the left-hand side of both equation gives $a = 2a$. The only value $a \in \mathbb{R}$ satisfying the equation is $a = 0$. Hence, the converse is true.

(d) Set $a = 0$ gives $U_0 = \left\{(x_1, x_2, x_3)^T \in \mathbb{R}^3 \middle| x_1 + 2x_2 = 0\right\}$. Define an operation $\oplus$ for the vectors in $\mathbb{R}^3$ by the rule:

$$\vec{u} \oplus \vec{v} = (u_1, u_2, u_3)^T \oplus (v_1, v_2, v_3)^T = (u_1 + v_1, u_2 - v_2, 0)^T$$

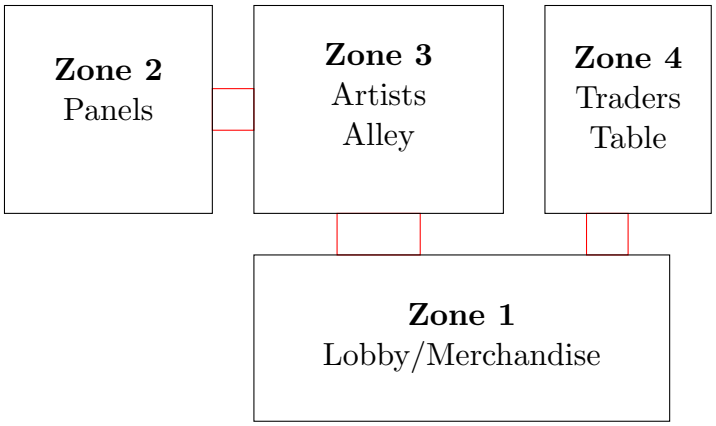
Suppose $\vec{u}, \vec{v} \in U_0$, show that $\vec{u} \oplus \vec{v} \in U_0$ if and only if $\vec{v}$ lies on the z-axis.

Solution: Since $\vec{u} = (u_1, u_2, u_3)^T, \vec{v} = (v_1, v_2, v_3)^T \in U_0$, this implies $u_1 + 2u_2 = 0$ and $v_1 + 2v_2 = 0$.
In one direction: If $\vec{v}$ lies on the z-axis, we can express $\vec{v} = (0, 0, c)^T$ for some $c \in \mathbb{R}$.
We have $\vec{u} \oplus \vec{v} = (u_1 + 0, u_2 - 0, 0)^T = (u_1, u_2, 0)^T \in U_0$.

In reverse direction: If $\vec{u} \oplus \vec{v} \in U_0$, this gives $(u_1 + v_1) + 2(u_2 - v_2) = 0$. Applying $u_1 + 2u_2 = 0$ and $v_1 + 2v_2 = 0$ gives: $-4v_2 = 0$.

Hence, v_2 must be zero, and this results in $v_1 = 0$ since $v_1 + 2(0) = 0$. As a result, $\vec{v} = (0, 0, c)^T$ for some $c \in \mathbb{R}$.

12 marks 4. A local comic book convention consists of four zones connected as follows:



Each day consists of sessions and attendees move between zones after between sessions with the following rules:

- 20% stays in the same zone
- The remaining moves to an adjacent zone with equal probability.

(a) 120 attendees participated on Day 1 of the convention. Determine the stable distribution of attendees. That is, the distribution where number of attendees in each zone remains unchanged at between sessions

Solution: Let x_i represents the number of guests in Zone i. Then, the question can be formulated as below:

$$\begin{array}{rrrrl} 0.2x_1 & & +0.4x_3 & +0.8x_4 & = x_1 \\ & 0.2x_2 & +0.4x_3 & & = x_2 \\ 0.4x_1 & +0.8x_2 & +0.2x_3 & & = x_3 \\ 0.4x_1 & & & +0.2x_4 & = x_4 \end{array}$$

Giving the augmented matrix below:

$$\left(\begin{array}{cccc|c} -0.8 & 0 & 0.4 & 0.8 & 0 \\ 0 & -0.8 & 0.4 & 0 & 0 \\ 0.4 & 0.8 & -0.8 & 0 & 0 \\ 0.4 & 0 & 0 & -0.8 & 0 \end{array}\right) \sim \left(\begin{array}{cccc|c} -0.8 & 0 & 0.4 & 0.8 & 0 \\ 0 & -0.8 & 0.4 & 0 & 0 \\ 0 & 0.8 & -0.6 & 0.4 & 0 \\ 0 & 0 & 0.2 & -0.4 & 0 \end{array}\right) \begin{array}{l} R_3 \to R_3 + \frac{1}{2}R_1 \\ R_4 \to R_4 + \frac{1}{2}R_1 \end{array}$$

$$\sim \left(\begin{array}{cccc|c} -0.8 & 0 & 0.4 & 0.8 & 0 \\ 0 & -0.8 & 0.4 & 0 & 0 \\ 0 & 0 & -0.2 & 0.4 & 0 \\ 0 & 0 & 0.2 & -0.4 & 0 \end{array}\right) \quad R_3 \to R_3 + R_2$$

$$\sim \left(\begin{array}{cccc|c} -0.8 & 0 & 0.4 & 0.8 & 0 \\ 0 & -0.8 & 0.4 & 0 & 0 \\ 0 & 0 & -0.2 & 0.4 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \quad R_4 \to R_4 + R_3$$

Let $x_4 = \alpha$, using back substitution:

$$x_3 = 2x_4 = 2\alpha$$
$$x_2 = \frac{1}{2}x_3 = \alpha$$
$$x_1 = \frac{1}{2}x_3 + x_4 = 2\alpha$$

Given $x_1 + x_2 + x_3 + x_4 = 120$, this yields $6\alpha = 120$ or $\alpha = 20$. The stable distribution is $(x_1, x_2, x_3, x_4)^T = (40, 20, 40, 20)^T$.

(b) Reports came in during the day about attendees breaking social distancing rules. As a result, a hard limit of 30 attendees are permitted in each zone during each session. Does there exist an stable distribution of 120 attendees that adheres to the new restrictions? Justify your answer.

Solution: For 120 attendees, the restrictions are not adhered to unless the initial distribution is $(x_1, x_2, x_3, x_4)^T = (30, 30, 30, 30)^T$. Applying the transition shows that $(30, 30, 30, 30)^T$ is not a stable distribution. Hence the distribution in next iteration must be different and there will exist $i \in \{1, 2, 3, 4\}$ such that $x_i > 30$. Hence, such initial distribution does not exist.

(c) Overnight, COVID-related incidents shut down the *Panels* and *Traders Tables* zones (Zones 2 and 4), leaving two zones open for the next day. We are tasked with predicting the range of potential behaviours of the remaining two zones.

Consider a general scenario with N attendees distributed over two zones as follows:

- α be the proportion of attendees who remain in zone 1 between sessions,
- β be the proportion of attendees who remain in zone 3 between sessions, and
- the remaining attendees switches zones.

Determine the stable distribution(s) in terms of N, α and β.
Note: You are not required to justify integer solutions in this context as number of convention attendees can be fuzzy with costumes and props etc.

Solution: Since α, β are proportions, we have $\alpha, \beta \in [0, 1]$. The question can be reformulated as below:

$$\alpha x_1 + (1 - \beta)x_3 = x_1$$
$$(1 - \alpha)x_1 + \beta x_3 = x_3$$

This gives the augmented matrix $\left(\begin{array}{cc|c} \alpha - 1 & 1 - \beta & 0 \\ 1 - \alpha & \beta - 1 & 0 \end{array}\right)$.

- **Generic case:** We can fix $\alpha - 1$ as a pivot entry:

$$\left(\begin{array}{cc|c} \alpha - 1 & 1 - \beta & 0 \\ 1 - \alpha & \beta - 1 & 0 \end{array}\right) \sim \left(\begin{array}{cc|c} \alpha - 1 & 1 - \beta & 0 \\ 0 & 0 & 0 \end{array}\right) \qquad R_2 \to R_2 + R_1$$

This results in $(\alpha - 1)x_1 + (1 - \beta)x_3 = 0$. As $x_1 + x_3 = N$, this gives:

$$(\alpha - 1)x_1 + (1 - \beta)(N - x_1) = 0$$

Solving the equation gives the stable distribution $\left(\frac{N(1-\beta)}{2-\alpha-\beta}, \frac{N(1-\alpha)}{2-\alpha-\beta}\right)^T$.

- Case $\alpha = 1$: The same computations apply as above, but the pivot entry is $1 - \beta$. This gives the stable distribution $(N, 0)^T$.
- Case $\alpha = 1$ and $\beta = 1$: The generic computation fails since the denominator gives zero. In this case, we have $\left(\begin{array}{cc|c} 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$. The solution set is $\mathbb{R}^2$. Hence, any distribution $(x_1, x_3)^T$ with $x_1 + x_3 = N$ is a stable distribution.